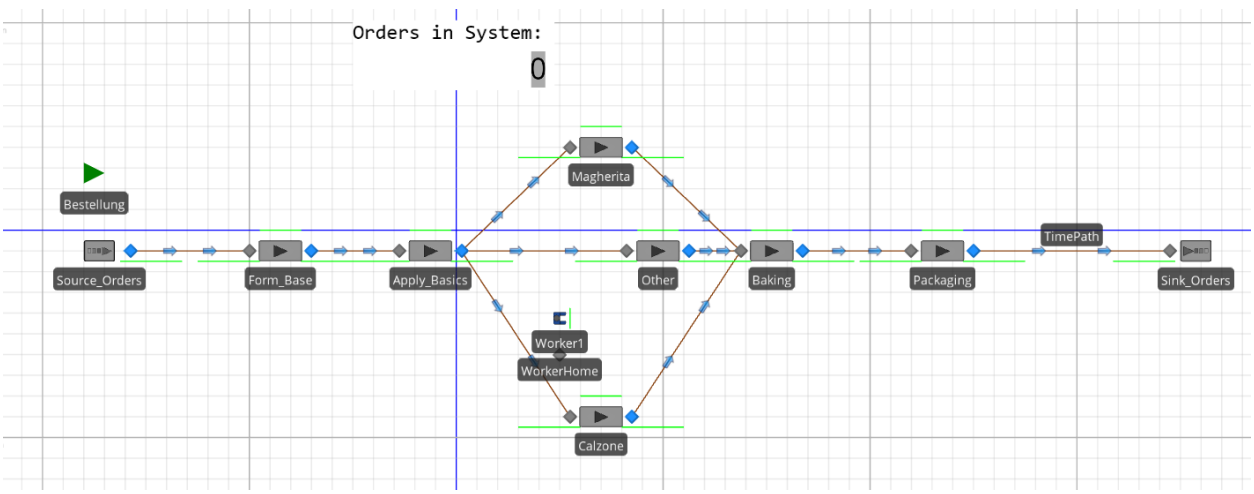
Assignment 2: Design of Experiments and Evaluation of a Given System

The Model consists of processes within a Pizza-Factory.
Formbase, Apply Basic Toppings, apply special Toppings, baking and packaging.

For each of these steps we assign different capacities for the experiments. E.g. Oven_capacity (How many Pizza fits in the oven) or FormBase_Capacity (How many dough can be formed at once, after an order has been entered).

Additionally, for the specialty topping we require staff(s), which are shared with 3 different stations.



The experiment planning Matrix:

Scenario				Replications		Controls										Responses		
	Name	Status	Last Run Comple...	Required	Completed	FormBase_Capacity	ApplyBasics_Capacity	Packaging_Capacity	Staff_Capacity	Oven_Cap...	Order_Rates	TopMagharita_Ca...	TopOther_Capa...	Calzone_Capacity	TimeInSystem_Mean	TimeInSystem_Max	Max_Waiting_Ord...
☑	Current	Completed	4/8/2026 3:37:3...	10	10 of 10	1	1	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.97047	5.82235	637.
☑	Scenario 1	Completed	4/8/2026 3:03:2...	10	10 of 10	2	1	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.64146	4.37137	108.
☑	Scenario 2	Completed	4/8/2026 3:03:2...	10	10 of 10	2	2	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.61107	4.3552	98.
☑	Scenario 3	Completed	4/8/2026 3:03:2...	10	10 of 10	2	3	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.61097	4.35462	98.
☑	Scenario 4	Completed	4/8/2026 3:03:2...	10	10 of 10	3	1	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.66237	4.45854	28.
☑	Scenario 5	Completed	4/8/2026 3:03:2...	10	10 of 10	3	2	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.60622	4.39329	28.
☑	Scenario 6	Completed	4/8/2026 3:03:2...	10	10 of 10	3	3	1	2	3	NormalRate	Infinity	Infinity	Infinity	1.61949	4.41788	33.

Variables

Controls								
FormBase_Capacity	ApplyBasics_Capacity	Packaging_Capacity	Staff_Capacity	Oven_Cap...	Order_Rates	TopMagharita_Ca...	TopOther_Capa...	Calzone_Capacity
2	1	1	2	3	NormalRate	Infinity	Infinity	Infinity
2	2	1	2	3	NormalRate	Infinity	Infinity	Infinity
2	3	1	2	3	NormalRate	Infinity	Infinity	Infinity
3	1	1	2	3	NormalRate	Infinity	Infinity	Infinity
3	2	1	2	3	NormalRate	Infinity	Infinity	Infinity
3	3	1	2	3	NormalRate	Infinity	Infinity	Infinity

1. Capacity: 4 different Stations
2. Staff

For the following experiments we have the variables **FormBase** and **ApplyBasics,** oven capacity, packaging capacity, and staff capacity (2). 3 Stations are not parameterized since they have fixed capacity.

We don't know which combination of the capacities and staff will give the optimum throughput. Therefore, we setup different scenarios with different capacities for our experiments.

KPIs

Responses		
TimeInSystem_Mean (Minutes)	TimeInSystem_Max (Minutes)	MaxNumberInSystem

Since we want to find the optimum throughput of the system, and find any bottlenecks, we measure the **time** an order **in the system** in **average** and **max** during the whole experiment. Additionally we want to measure the **maximum number of (simultaneous) orders within the system** to get another (more deterministic) indication. Of course taking the max is only considering the peak-demand time, but this is sufficient for our purpose.

Implementation

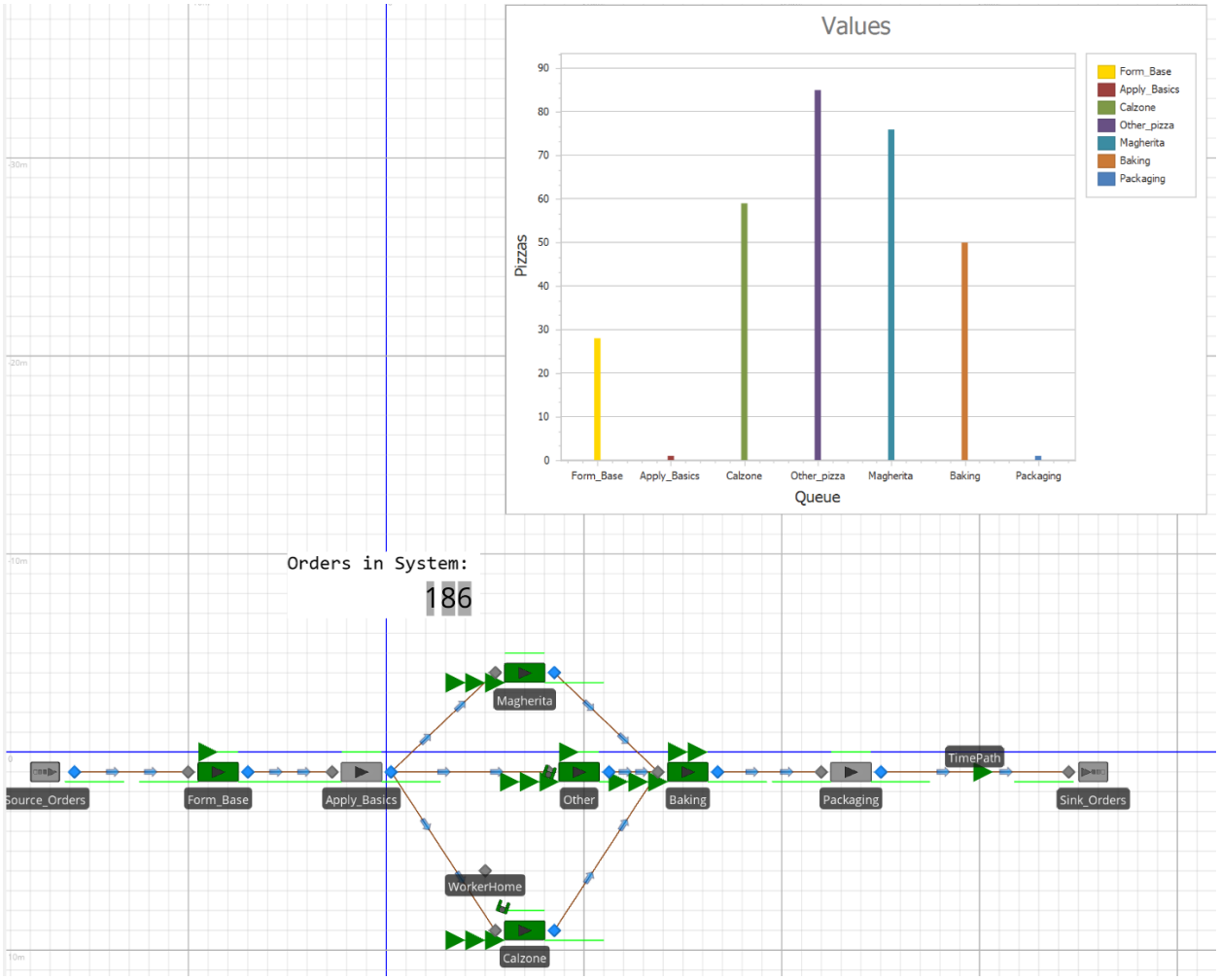
Scenario				Replications		Controls	
☑	Name	Status	Last Run Comple...	Required	Completed	FormBase_Capacity	ApplyBasics_Capacity
☑	Current	Completed	4/8/2026 4:01:1...	10	10 of 10	1	1
☑	Scenario1	Completed	4/8/2026 4:01:1...	10	10 of 10	2	1
☑	Scenario2	Completed	4/8/2026 4:01:1...	10	10 of 10	2	2
☑	Scenario3	Completed	4/8/2026 4:01:1...	10	10 of 10	3	1
☑	Scenario4	Completed	4/8/2026 4:01:1...	10	10 of 10	3	2
☑	Scenario5	Completed	4/8/2026 4:01:1...	10	10 of 10	3	3

For the experiments we setup 1. the current scenario, to confirm and validate the parameters with the real world.

In Scenario 1 to 5, we find out what happens if we add different amount of capacity to FormBase and ApplyBasics, but we never add more capacity to ApplyBasics than FormBase, because the time need for ApplyBasics is less than FormBase in most of the time.

Scenario				Replications		Controls									Responses		
☑	Name	Status	Last Run Comple...	Required	Completed	FormBase_Capacity	ApplyBasics_Capacity	...	...	...	...	...	...	...	TimeInSystem_Mean	TimeInSystem_Max	Max_Waiting_Ord...
☑	Current	Completed	4/8/2026 4:01:1...	10	10 of 10	1	1	1	2	3	...	...	...	I...	1.6827	4.61745	247.1
☑	Scenario1	Completed	4/8/2026 4:01:1...	10	10 of 10	2	1	1	2	3	...	...	...	I...	1.64146	4.37137	238.6
☑	Scenario2	Completed	4/8/2026 4:01:1...	10	10 of 10	2	2	1	2	3	...	...	...	I...	1.61107	4.3552	226.9
☑	Scenario3	Completed	4/8/2026 4:01:1...	10	10 of 10	3	1	1	2	3	...	...	...	I...	1.66237	4.45854	239.9
☑	Scenario4	Completed	4/8/2026 4:01:1...	10	10 of 10	3	2	1	2	3	...	...	...	I...	1.60622	4.39329	238.1
☑	Scenario5	Completed	4/8/2026 4:01:1...	10	10 of 10	3	3	1	2	3	...	...	...	I...	1.61949	4.41788	242.1

As we can see, there is minimal improvement in the overall performance / throughput after increasing the capacity of Formbase and ApplyBasics. In terms of additional expenses scenario 2 and all the following scenarios does not significant decrease the max_waiting_order or the timeInSystem_Max. From that we can also derive that the bottle neck has basically moved to subsequent steps (apply specialty toppings and baking).



To confirm it we run the experiment with the Form_base capacity 2 again and see, that indeed the bottle-neck has moved to apply specialty toppings. In this step 2 staff are sharing 3 stations. As we have only 3 station with fixed capacity. This is also the limiting factor in the whole system, and adding more Staff, does not help in this case.

Mathematically, if we disregard the randomness of timing of the stations and just take the maximum, we can almost use the common denominator of order per minute for each station to find the perfect ratio, because this model is a almost linear sequence, except for the specialty toppings. But there we can again use the average of all station since the link weight are the same.

3x1.2x((6+1)+5/3)x2.333x2 = 67.2

- ➔ Form base 3/67.2*100 =4.46 (*1.12 for rounding) -> 5
- ➔ Apply basic 1.2/67.2*100 =1.78 ->2
- ➔ Staff 4/67.2*100 = 5.95 ->6 (maximum 3 stations currently -> 3)
- ➔ Baking 2.33/67.2*100 = 3.47 -> 4
- ➔ Packaging 2/67.2*100 =2.98 ->3

Results

Design | Response Results | Pivot Grid | Reports | Dashboard Reports | Input Analysis

Scenario				Replications		Controls									Responses		
	Name	Status	Last Run Comple...	Required	Completed	FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Oven_Capacity	Packaging_Cap...	Orde...	...	...	...	TimeInSystem_Mean (...	TimeInSystem_Max (Min...	MaxNumberIn...
☑	Current	Completed	4/8/2026 8:16:3...	10	10 of 10	1	1	2	3	1	Nor...	...	...	...	136.08	355.746	246.5
☑	Scenario1	Completed	4/8/2026 8:16:3...	10	10 of 10	2	1	2	3	1	Nor...	...	...	...	132.044	346.306	237.9
☑	Scenario2	Completed	4/8/2026 8:16:3...	10	10 of 10	2	2	2	3	1	Nor...	...	...	...	129.445	341.274	226.3
☑	Scenario3	Completed	4/8/2026 8:16:3...	10	10 of 10	5	2	3	4	3	Nor...	...	...	...	120.251	311.156	208
☑	Scenario4	Completed	4/8/2026 8:16:3...	10	10 of 10	10	2	3	4	3	Nor...	...	...	...	122.995	316.665	218.9
☑	Scenario5	Completed	4/8/2026 8:16:3...	10	10 of 10	5	10	3	4	3	Nor...	...	...	...	119.838	311.956	210.3
☑	Scenario6	Completed	4/8/2026 8:16:3...	10	10 of 10	5	2	10	4	3	Nor...	...	...	...	120.251	311.156	208
☑	Scenario7	Completed	4/8/2026 8:16:3...	10	10 of 10	5	2	3	10	3	Nor...	...	...	...	119.681	311.021	211
☑	Scenario8	Completed	4/8/2026 8:16:3...	10	10 of 10	5	2	3	4	10	Nor...	...	...	...	119.259	309.986	208.1

As we can see in Scenario 3, after we applied the formula we have maximum simultaneous orders in system of 208, which is the lowest until now.

To confirm that this is also the optimum, we increase the capacity by 10 in the following scenarios (5-8) for each variable separately, and see that the result is either worse or the same.

Conclusion

Since perfect equilibrium of a system is very difficult to achieve if humans are involved, and therefore the bottle neck will anyways move from one place to another anyways. But given the limitation of 3 stations with fixed capacity in the system, we were still able to find the optimum using the perfect ratio.

Since the owner of the factory, does not want to change the capacity of the existing specialty topping stations, but rather add other pizza styles, we suggest the following for improvement:

Work towards the capacity ratios of the different station in the system as following: 5 Form_base, 2 Apply_basics, 3 Specialty Topping, 4 Baking, and 3 Packaging.
And also, it is possible to add 3 additional Pizza style, so that customer has more choice, but also, this would improve the overall system throughput. We can reach a 15% reduction in wait time. Due to the fact that employing another staff, also increase the bottom line cost, the owner should consider only hiring them for the evenings, booming hours.